



Dhirubhai Ambani Institute of Information and
Communication Technology (DA-IICT)

Discrete Mathematics SC612

IN SEMESTER 2 EXAMINATION

Max Marks - 50

Autumn 2023-24

Time - 1:30 hours

- All questions are compulsory.
- Answers must have proper justification to get full credit.
- Understanding the questions is part of the evaluation. Don't ask for any clarification. In case of ambiguity, proceed with an assumption by clearly stating it.

1. Solve the recurrence relation using characteristic roots method

[10]

$$a_n = 4a_{n-1} - 3a_{n-2} + 2^n + n + 3^n + 5 + n^2$$

Solution: To solve the recurrence relation $a_n = 4a_{n-1} - 3a_{n-2} + F(n)$, we'll first find the characteristic roots of the homogeneous part of the relation, and then find a particular solution for the non-homogeneous part. Finally, we'll combine the solutions to get the general solution.

Homogeneous Part: The homogeneous part of the recurrence relation is $a_n = 4a_{n-1} - 3a_{n-2}$. To solve this homogeneous relation, we assume a solution of the form $a_n = r^n$ and substitute it into the equation:

$$r^n = 4r^{n-1} - 3r^{n-2}$$

Divide the entire equation by r^{n-2} (assuming $r \neq 0$):

$$r^2 = 4r - 3$$

This is a quadratic equation in r , which can be factored:

$$r^2 - 4r + 3 = (r - 1)(r - 3) = 0$$

So, we have two characteristic roots: $r_1 = 1$ and $r_2 = 3$.

Non-Homogeneous Part: The non-homogeneous part of the recurrence relation is $F(n) = 2^n + 3^n + n^2 + n + 5$. To find a particular solution for this part, we make an educated guess based on the form of the terms. Since the non-homogeneous part contains terms like 2^n , 3^n , 1^n and the bases 1 and 3 are characteristic roots, we assume a particular solution of the form:

$$a_n^{(p)} = A \cdot 2^n + Bn \cdot 3^n + n(C \cdot n^2 + D \cdot n + E)1^n$$

where A , B , C , D , and E are constants to be determined.

Now, substitute this particular solution into the original recurrence relation:

$$\begin{aligned} & A \cdot 2^n + Bn \cdot 3^n + n(C \cdot n^2 + D \cdot n + E) \\ &= 4(A \cdot 2^{n-1} + B(n-1) \cdot 3^{n-1} + (n-1)(C \cdot (n-1)^2 + D \cdot (n-1) + E)) - \\ & \quad 3(A \cdot 2^{n-2} + B(n-2) \cdot 3^{n-2} + (n-2)(C \cdot (n-2)^2 + D \cdot (n-2) + E)) + 2^n + 3^n + n^2 + n + 5 \\ &= [8A - 3A + 4] \cdot 2^{n-2} + [4B(n-1) - 3B(n-2) + 9] \cdot 3^{n-2} + \\ & \quad 4(n-1)(C \cdot (n-1)^2 + D \cdot (n-1) + E) - 3(n-2)(C \cdot (n-2)^2 + D \cdot (n-2) + E) + n^2 + n + 5 \\ &= [5A + 4] \cdot 2^{n-2} + [Bn + 11] \cdot 3^{n-2} + 4(n-1)(Cn^2 - 2Cn + C + Dn - D + E) \\ & \quad - 3(n-2)(Cn^2 - 4Cn + 4C + Dn - 2D + E) + n^2 + n + 5 \\ &= [5A + 4] \cdot 2^{n-2} + [Bn + 11] \cdot 3^{n-2} + 4(Cn^3 - 2Cn^2 + Cn + Dn^2 - Dn + En - Cn^2 + 2Cn \\ & \quad - C - Dn + D - E) - 3(Cn^3 - 4Cn^2 + 4Cn + Dn^2 - 2Dn + En - 2Cn^2 + 8Cn - 8C \\ & \quad - 2Dn + 4D - 2E) + n^2 + n + 5 \\ &= [5A + 4] \cdot 2^{n-2} + [Bn + 11] \cdot 3^{n-2} + Cn^3 + 4[(D - 3C)n^2 + (-3C - 2D + E)n - C + D - E] \\ & \quad - 3[(-6C + D)n^2 + (12C - 4D + E)n - 8C + 4D - 2E] + n^2 + n + 5 \\ &= [5A + 4] \cdot 2^{n-2} + [Bn + 11] \cdot 3^{n-2} + Cn^3 + (D + 6C + 1)n^2 + (-48C + 4D + E + 1)n \\ & \quad + 20C - 8D + 2E + 5 \end{aligned}$$

General Solution: The general solution is the sum of the homogeneous and particular solutions:

$$a_n = a_n^{(h)} + a_n^{(p)}$$

The homogeneous solution is a linear combination of the characteristic roots:

$$a_n^{(h)} = c_1 \cdot r_1^n + c_2 \cdot r_2^n$$

where c_1 and c_2 are constants to be determined from initial conditions.

So, the final general solution is:

$$a_n = c_1 \cdot 1^n + c_2 \cdot 3^n + A \cdot 2^n + Bn \cdot 3^n + n(C \cdot n^2 + D \cdot n + E)1^n$$

where c_1 and c_2 are determined by the initial conditions for a_0 and a_1 .

2. (a) How many numbers greater than 10^6 can be formed using the digits 1, 2, 0, 2, 4, 2, 4?

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Solution: To form numbers greater than 10^6 using the given digits 1, 2, 0, 2, 4, 2, and 4, we need to consider only 1, 2, or 4 as the first digit:

Numbers starting with 1: 1 is fixed as the first digit.

There are three 2's, one 0, and two 4's remaining to choose from for the second, third, fourth, fifth, and sixth digits

So, there are $\frac{6!}{3!2!} = 60$ numbers that start with 1.

Numbers starting with 2: 2 is fixed as the first digit.

There are 0, 1, two 2's, and two 4's remaining to choose from for the second, third, fourth, fifth, and sixth digits

So, there are a total of $\frac{6!}{2!2!} = 180$ numbers that start with 2.

Numbers starting with 4: 4 is fixed as the first digit.

There are 0, 1, three 2's, and 4 remaining to choose from for the second, third, fourth, fifth, and sixth digits

So, there are a total of $\frac{6!}{3!} = 120$ numbers that start with 4.

Total Now, sum up the numbers of each case: $60 + 180 + 120 = 360$

So, there are 360 numbers greater than 10^6 that can be formed using the given digits.

- (b) The Lucas sequence is defined by $L_n = L_{n-1} + L_{n-2}$, for $n > 2$ with $L_1 = 1$ and $L_2 = 3$. Find L_n .

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Solution: The given recurrence relation is $L_n = L_{n-1} + L_{n-2}$

$$\therefore L_n - L_{n-1} - L_{n-2} = 0$$

Therefore, the characteristic equation is $r^2 - r - 1 = 0$

Solving for r gives $r_1 = \frac{1+\sqrt{5}}{2}$, $r_2 = \frac{1-\sqrt{5}}{2}$

Therefore, the general solution is of the form $L_n = \alpha (r_1)^n + \beta (r_2)^n$

$$\therefore L_n = \alpha \left(\frac{1+\sqrt{5}}{2} \right)^n + \beta \left(\frac{1-\sqrt{5}}{2} \right)^n$$

Now, given $L_1 = 1$

$$\therefore L_1 = 1 = \alpha \left(\frac{1 + \sqrt{5}}{2} \right) + \beta \left(\frac{1 - \sqrt{5}}{2} \right) = (\alpha + \beta) \frac{1}{2} + (\alpha - \beta) \frac{\sqrt{5}}{2}$$

$$\therefore \alpha + \beta = 2 \text{ and } \alpha - \beta = 0$$

$$\therefore \alpha = 1, \beta = 1$$

$$\therefore L_n = \left(\frac{1 + \sqrt{5}}{2} \right)^n + \left(\frac{1 - \sqrt{5}}{2} \right)^n$$

3. Solve the recurrence relation using generating function method

[10]

$$a_n - 2a_{n-1} + 2a_{n-2} - a_{n-3} = 0$$

with initial conditions $a_0 = 2$ and $a_1 = 1 = a_2$.

Solution: To solve this recurrence relation using generating functions, we'll define the generating function $G(x)$ as follows:

$$G(x) = \sum_{n=0}^{\infty} a_n x^n$$

Now, we'll manipulate the given recurrence relation to obtain an equation involving $G(x)$. The recurrence relation is:

$$a_n - 2a_{n-1} + 2a_{n-2} - a_{n-3} = 0$$

Multiplying both sides by x^n and summing over n , we get:

$$\sum_{n=3}^{\infty} (a_n - 2a_{n-1} + 2a_{n-2} - a_{n-3}) x^n = 0$$

Now, let's express each term using the generating function $G(x)$:

$$\sum_{n=3}^{\infty} a_n x^n - 2 \sum_{n=3}^{\infty} a_{n-1} x^n + 2 \sum_{n=3}^{\infty} a_{n-2} x^n - \sum_{n=3}^{\infty} a_{n-3} x^n = 0$$

We can shift the indices in the sums as follows:

$$\begin{aligned}
\sum_{n=3}^{\infty} a_n x^n - 2x \sum_{n=3}^{\infty} a_{n-1} x^{n-1} + 2x^2 \sum_{n=3}^{\infty} a_{n-2} x^{n-2} - x^3 \sum_{n=3}^{\infty} a_{n-3} x^{n-3} &= 0 \\
\sum_{n=3}^{\infty} a_n x^n - 2x \sum_{n=2}^{\infty} a_n x^n + 2x^2 \sum_{n=1}^{\infty} a_n x^n - x^3 \sum_{n=0}^{\infty} a_n x^n &= 0 \\
[G(x) - a_0 - a_1 x - a_2 x^2] - 2x[G(x) - a_0 - a_1 x] + 2x^2[G(x) - a_0] - x^3 G(x) &= 0 \\
G(x)[x^3 - 2x^2 + 2x - 1] &= -a_0 - a_1 x - a_2 x^2 - 2x(-a_0 - a_1 x) - 2x^2 a_0
\end{aligned}$$

Using the initial conditions on the RHS:

$$\begin{aligned}
G(x)[x^3 - 2x^2 + 2x - 1] &= -2 - x - x^2 - 2x(-2 - x) - 4x^2 = -2 + 3x - 3x^2 \\
G(x) &= \frac{-2 + 3x - 3x^2}{x^3 - 2x^2 + 2x - 1} = \frac{-2 + 3x - 3x^2}{(x-1)(x^2 - x - 1)}
\end{aligned}$$

We can now decompose the rational expression as follows:

$$\begin{aligned}
\frac{-2 + 3x - 3x^2}{(x-1)(x^2 - x - 1)} &= \frac{A}{x-1} + \frac{B}{x - \frac{1+\sqrt{5}}{2}} + \frac{C}{x - \frac{1-\sqrt{5}}{2}} \\
-2 + 3x - 3x^2 &= A(x^2 - x - 1) + B(x-1) \left(x - \frac{1-\sqrt{5}}{2} \right) + C(x-1) \left(x - \frac{1+\sqrt{5}}{2} \right)
\end{aligned}$$

Take $x = 1$, we get $-2 = -A \implies A = 2$

$$\begin{aligned}
\text{Take } x = \frac{1-\sqrt{5}}{2}, \text{ we get } -2 + 3(-1) &= C \left(\frac{1-\sqrt{5}}{2} - 1 \right) \left(\frac{1-\sqrt{5}}{2} - \frac{1+\sqrt{5}}{2} \right) \\
\implies -5 = \sqrt{5} C \frac{1+\sqrt{5}}{2} &\implies C = -\frac{1+\sqrt{5}}{2\sqrt{5}}
\end{aligned}$$

$$\begin{aligned}
\text{Take } x = \frac{1+\sqrt{5}}{2}, \text{ we get } -2 + 3(-1) &= B \left(\frac{1+\sqrt{5}}{2} - 1 \right) \left(\frac{1+\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2} \right) \\
\implies -5 = \sqrt{5} B \frac{-1+\sqrt{5}}{2} &\implies B = \frac{1-\sqrt{5}}{2\sqrt{5}}
\end{aligned}$$

Therefore,

$$\begin{aligned}
G(x) &= \frac{A}{x-1} + \frac{B}{x - \frac{1+\sqrt{5}}{2}} + \frac{C}{x - \frac{1-\sqrt{5}}{2}} \\
&= \frac{-A}{1-x} + \frac{-B}{\frac{1+\sqrt{5}}{2} \left[1 - \frac{2}{1+\sqrt{5}} x \right]} + \frac{-C}{\frac{1-\sqrt{5}}{2} \left[1 - \frac{2}{1-\sqrt{5}} x \right]} \\
G(x) &= \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} -A x^n + \frac{-B}{\frac{1+\sqrt{5}}{2}} \sum_{n=0}^{\infty} \left[\frac{2}{1+\sqrt{5}} \right]^n x^n + \frac{-C}{\frac{1-\sqrt{5}}{2}} \sum_{n=0}^{\infty} \left[\frac{2}{1-\sqrt{5}} \right]^n x^n \\
\implies a_n &= -A + \frac{-B}{\frac{1+\sqrt{5}}{2}} \left[\frac{2}{1+\sqrt{5}} \right]^n + \frac{-C}{\frac{1-\sqrt{5}}{2}} \left[\frac{2}{1-\sqrt{5}} \right]^n
\end{aligned}$$

4. The English alphabet contains 21 consonants and five vowels. How many strings of six lowercase letters of the English alphabet contain

[10]

- (a) exactly one vowel?
- (b) exactly two vowels?
- (c) at least one vowel?
- (d) at least two vowels?

Solution: With repetition

There are 5 vowels and 21 consonants in the English alphabet. We need to be careful here, because strings can have repeated letters.

- a) To find the number of strings with exactly one vowel, we can consider the following:
- 1. We need to choose the position for the vowel: this can be done in 6 ways.
 - 2. We need to choose the vowel to use: this can be done in 5 ways.
 - 3. Each of the other five positions in the string can contain any of the 21 consonants: so there are 21^5 ways to fill the rest of the string.

Therefore the answer is $6 \cdot 5 \cdot 21^5 = 122,523,030$.

- b) To find the number of strings with exactly two vowels, we can consider the following:
- 1. We need to choose the position for the vowels: this can be done in $C(6, 2) = 15$ ways (we need to choose two positions out of six).
 - 2. We need to choose the two vowels (5^2 ways).
 - 3. Each of the other four positions in the string can contain any of the 21 consonants, so there are 21^4 ways to fill the rest of the string.

Therefore, the answer is $15 \cdot 5^2 \cdot 21^4 = 72,930,375$.

- c) To find the number of strings with at least one vowel, we need to consider the following cases:
- 1. count the number of strings without vowels: 21^6 ($\because$ 6 positions for 21 consonants)
 - 2. subtract the above from the total number of strings: 26^6 .

Therefore, the answer is $26^6 - 21^6 = 223,149,655$.

- d) To find the number of strings with at least two vowels, we will do the following:
- 1. find the total number of strings: 26^6
 - 2. find the number of strings with no vowels: 21^6
 - 3. find the number of strings with one vowel (computed in part (a))

Therefore, the answer is $26^6 - 21^6 - 6 \cdot 5 \cdot 21^5 = 223149655 - 122523030 = 100,626,625$.

5. (a) Devise a recursive algorithm to find the n^{th} term of the sequence defined by

[5]

$$a_0 = 1, a_1 = 2, \text{ and } a_n = a_{n-1} \cdot a_{n-2}, \text{ for } n = 2, 3, 4, \dots$$

Solution: You can devise a recursive algorithm to find the n th term of the sequence defined by $a_0 = 1$, $a_1 = 2$, and $a_n = a_{n-1} \cdot a_{n-2}$ for $n = 2, 3, 4, \dots$. Here's a Python implementation of the algorithm:

Algorithm 1: Recursive Algorithm to Find the n -th Term of a Sequence

Input : A positive integer n
Output: The n -th term of the sequence
Data: $a_0 = 1$, $a_1 = 2$
Result: The n -th term a_n
if $n = 0$ **then**
 | **return** 1;
if $n = 1$ **then**
 | **return** 2;
 $a_{n-2} \leftarrow \text{find_nth_term}(n - 2);$
 $a_{n-1} \leftarrow \text{find_nth_term}(n - 1);$
return $a_{n-1} \times a_{n-2};$

This recursive algorithm first checks for the base cases when n is 0 or 1. For these cases, it returns the corresponding values of a_0 and a_1 . For values of n greater than 1, it calculates the n th term using the recursive formula $a_n = a_{n-1} \cdot a_{n-2}$.

- (b) Verify that the program segment

[5]

$$\begin{aligned} & \text{if } x < y \text{ then} \\ & \quad \text{min} := x \\ & \text{else} \\ & \quad \text{min} := y \end{aligned}$$

is correct with respect to the initial assertion T and the final assertion $(x \leq y \wedge \text{min} = x) \vee (x > y \wedge \text{min} = y)$.

Solution: The program checks if x is less than y . Depending on the result of this comparison, it assigns the value of x or y to min .

After the execution of the program, one of the following conditions holds:

- a) If $x < y$, the statement $\text{min} := x$ is executed.

In this case, we have:

$x < y$ (from the initial comparison) and $\text{min} = x$ (since min is assigned the value of x).

final assertion: $(x \leq y \wedge \text{min} = x) \vee (x > y \wedge \text{min} = y) = (T \wedge T) \vee (F \wedge F) = T \vee F = T$

b) If $x > y$, the statement $\text{min} := y$ is executed.

In this case, we have:

$x > y$ (from the initial comparison) and $\text{min} = y$ (since min is assigned the value of y)

final assertion: $(x \leq y \wedge \text{min} = x) \vee (x > y \wedge \text{min} = y) = (F \wedge F) \vee (T \wedge T) = F \vee T = T$

c) If $x = y$, the statement $\text{min} := y$ is executed.

In this case, we have:

$x = y$ (from the initial comparison) and $\text{min} = y$ (since min is assigned the value of y).

final assertion: $(x \leq y \wedge \text{min} = x) \vee (x > y \wedge \text{min} = y) = (T \wedge T) \vee (F \wedge T) = T \vee F = T$

In all the three cases, the final assertion $(x \leq y \wedge \text{min} = x) \vee (x > y \wedge \text{min} = y)$ holds true.

The program behaves correctly in assigning the minimum value between x and y to min based on the comparison result.

Therefore, the given program segment is correct with respect to the initial assertion and the final assertion.

***** END *****

This exam has 5 questions, for a total of 50 marks.

Question:	1	2	3	4	5	Total
Points:	10	12	10	8	10	50
Score:						